Logarithm

Mind map

1. Exponential and log form

- $N = a^x \iff \log_a N = x$
- $N > 0,\ a > 0,\ a \neq 1$
- same base: powers equal

2. Basic values

- $\log_a a = 1,\ \log_a 1 = 0$
- $\log_a \frac{1}{a} = -1$
- $a^{\log_a N} = N$

3. Properties

- sum $\to$ product, difference $\to$ quotient
- $\log_a m^p = p \log_a m$
- $\log_{a^p} m = \frac{1}{p} \log_a m$

4. Base change

- $\log_a b = \frac{\log_c b}{\log_c a}$
- $\log_a b = 1/\log_b a$
- $a^{\log_c b} = b^{\log_c a}$
- sign: same side of $1 \Rightarrow +$

5. Characteristic, digits, zeroes

- $\log_{10} N = c + m,\ 0 \leq m < 1$
- digits $= c + 1$
- zeroes after decimal $= |c + 1|$

6. Log equations

- remove log, check domain
- repeated log: $t = \log_a x$
- exponent: $t = a^x > 0$

7. Variable power

- $f^g = h$: take log
- $a^b = a^c \Rightarrow a = 1$ or $b = c$

8. Log inequalities

- base > 1: same direction
- base in $(0, 1)$: reverse
- number > 0 always

9. Modulus basics

- $|x| = x\ (x \geq 0),\ -x\ (x < 0)$
- $\sqrt{x^2} = |x|,\ |x|^2 = x^2$
- $|x| \geq 0$

10. Modulus equations

- $|f| = k \Rightarrow f = \pm k\ (k \geq 0)$
- $|f| = |g| \Rightarrow f^2 = g^2$
- else define by intervals

11. Modulus inequalities

- $|f| < k \iff -k < f < k$
- $|f| > k \iff f > k$ or $f < -k$
- $|f| < |g| \iff f^2 < g^2$

12. Exponential inequalities

- same base first
- $a > 1$: same direction; $0 < a < 1$: reverse

13. Irrational inequalities

- $\sqrt{f} < g$: $f \geq 0,\ g > 0,\ f < g^2$
- $\sqrt{f} > g$: two cases, union

14. Comparing and radicals

- compare with 1 or same base
- $\sqrt{a \pm 2\sqrt{b}} = \sqrt{p} \pm \sqrt{q}$